UC SAN DIEGO / ECE 251C

6

Generalized sampling and multiresolution representations

Executive summary

Generalized sampling replaces point evaluations by measurements against an acquisition function. With a box-shaped acquisition function, these measurements produce local averages and piecewise-constant approximations. We normalize the boxes, compare adjacent resolutions, and identify their difference with Haar wavelets. This leads to approximation spaces V_i, detail spaces W_i, and the orthogonal decomposition $V_{i+1} = V_i \oplus W_i$. The two-scale equations give filters that convert between fine coefficients and coarse coefficients plus details. Finally, we examine the fine- and coarse-resolution limits that make this construction a multiresolution analysis of $L^2(\mathbb{R})$.

6.1 Generalized sampling

In Lecture 5, projecting onto normalized boxes replaced a signal by its average on each cell. We now interpret these averages as measurements supplied by an acquisition system. This connects the continuous-domain signal to the discrete coefficients that we can store and process. Throughout this lecture the domain is the whole line $\mathbb{R}$; no periodic extension is imposed.

Recall: ideal sampling. With unit sampling step,

$$f[n] = f(n) = \langle f, \delta(\cdot - n)\rangle, \qquad n \in \mathbb{Z}.$$

The impulse pairing means evaluation at n, whenever that value is defined; it is not an L^2 inner product with δ.

Generalized sampling. For $f \in L^2(\mathbb{R})$, replace the impulse by a real-valued acquisition function $\phi \in L^2(\mathbb{R})$:

$$a[n] \coloneqq \langle f, \phi(\cdot - n)\rangle_{L^2} = \int_{\mathbb{R}} f(t)\phi(t-n)\,\mathrm{d}t. \tag{6.1}$$

The second function is conjugated in the inner product; here $\phi^* = \phi$ because the acquisition function is real-valued. Cauchy–Schwarz guarantees that each measurement exists for $f, \phi \in L^2(\mathbb{R})$. Unlike ideal sampling, the measurement

combines values of the signal over the region where the acquisition function is nonzero.

Filtering before sampling. Set $\widetilde{\phi}(t) \coloneqq \phi(-t)$. Then

$$a[n] = \int_{\mathbb{R}} f(t)\widetilde{\phi}(n-t)\,\mathrm{d}t = (f * \widetilde{\phi})(n).$$

Thus generalized sampling can be implemented by filtering with impulse response $\widetilde{\phi}$ and then sampling the output. The reflection is required because an inner product contains $\phi(t-n)$, whereas convolution uses a kernel evaluated at $n-t$. For a complex-valued acquisition function, the corresponding kernel would be $\widetilde{\phi}(t) = \phi^*(-t)$.

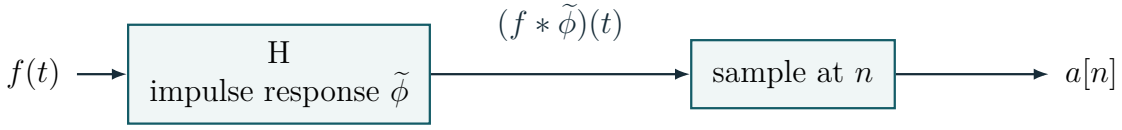


Figure 6.1: Generalized sampling is filtering followed by ideal sampling. The reflection in $\widetilde{\phi}(t) = \phi(-t)$ converts the inner-product formula into convolution.

6.2 Box measurements and piecewise-constant reconstruction

The simplest acquisition function is a rectangular pulse, or unit box:

$$\phi(t) = \mathbf{1}_{[0,1)}(t), \qquad \widetilde{\phi}(t) = \phi(-t).$$

The choice of values at the endpoints does not affect an integral.

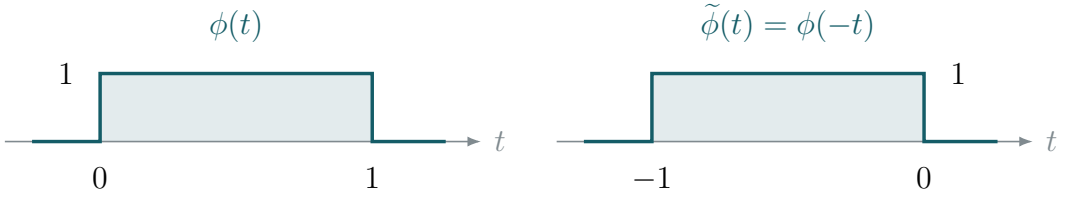


Figure 6.2: The acquisition box and its reflected impulse response. At output location n, convolution with the reflected box integrates f over $[n, n+1)$.

For this choice, equation (6.1) becomes

$$a[n] = \int_{n}^{n+1} f(t)\,\mathrm{d}t. \tag{6.2}$$

Since the interval has length one, this integral equals the average of f on the interval. The translates $\phi(t-n)$ have disjoint interiors and each has energy one, so they form an orthonormal family. This special property will allow the acquisition coefficients to be used directly as coefficients of an orthogonal projection.

To obtain an approximation from the measurements, place each measured value on the interval from which it was acquired. In terms of the same translated boxes, this gives

$$f_0(t) \coloneqq \sum_{n\in\mathbb{Z}} a[n]\phi(t-n). \tag{6.3}$$

For $n \le t < n+1$, only the nth term is nonzero. Thus $f_0(t) = a[n]$: we replace each unit interval by its average value.

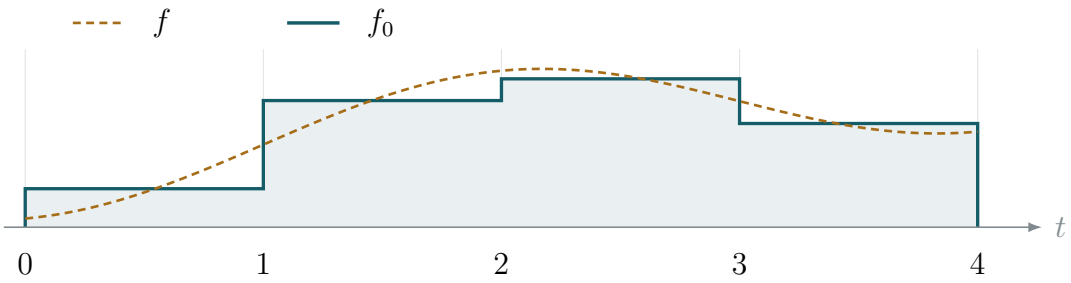


Figure 6.3: Four distinct unit-cell averages of $f(t) = 0.7 + 0.25t - 0.6\cos(\pi t/2)$ on $[0, 4)$, with zero values outside that interval. Each box height is an average, not a point sample.

The acquisition and reconstruction functions play different roles. The reflected box forms a measurement by convolution, while the unreflected box places that measurement on $[n, n+1)$. The approximation reproduces every signal that is constant on each unit interval, but not an arbitrary signal: variation within a cell has been replaced by one average. For a general acquisition function, using the same translates for reconstruction need not produce an orthogonal projection.

6.3 Changing resolution without changing energy

To double the resolution, replace $\phi(t-n)$ by $\phi(2t-n)$. Its nonzero interval has half the width. More generally, for an integer resolution i, define

$$I_{i,n} \coloneqq [n2^{-i}, (n+1)2^{-i}), \qquad n \in \mathbb{Z}.$$

The function $\phi(2^i t - n)$ occupies this cell, whose width is 2^{-i}. Increasing i means finer resolution; negative i gives cells wider than one.

The amplitude needs a normalization. The unnormalized box has energy

$$\int_{\mathbb{R}} |\phi(2^i t - n)|^2 \, \mathrm{d}t = 2^{-i} \int_{\mathbb{R}} |\phi(u)|^2 \, \mathrm{d}u = 2^{-i}, \qquad u = 2^i t - n.$$

Its energy shrinks with its support and tends to zero as $i \to \infty$. For fixed n, the support also contracts toward the origin, so the unnormalized box eventually

vanishes at every fixed $t \neq 0$. To compare basis coefficients across resolutions, normalize all sampling functions to unit energy:

$$\boxed{\phi_{i,n}(t) := 2^{i/2}\phi(2^i t - n), \qquad i, n \in \mathbb{Z}.} \tag{6.4}$$

With $u = 2^i t - n$ and $\mathrm{d}t = 2^{-i}\mathrm{d}u$,

$$\begin{aligned} \|\phi_{i,n}\|_{L^2}^2 &= 2^i \int_{\mathbb{R}} |\phi(2^i t - n)|^2 \,\mathrm{d}t \\ &= 2^i 2^{-i} \int_{\mathbb{R}} |\phi(u)|^2 \,\mathrm{d}u = 1. \end{aligned}$$

The factor $2^{i/2}$ therefore compensates exactly for the change in width. The same calculation works for any unit-energy function, not only a box. It is the normalization used throughout the wavelet construction.

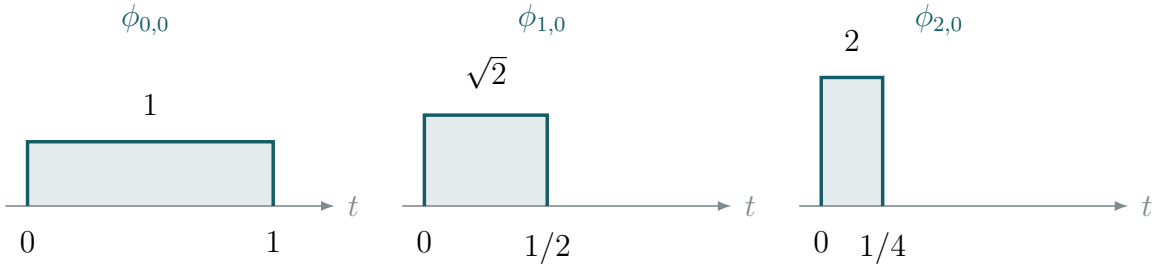


Figure 6.4: Unit-energy boxes at three resolutions, drawn with common horizontal and vertical units. Halving the width multiplies the height by $\sqrt{2}$, keeping height squared times width equal to one.

Unit *energy* should not be confused with unit *area*. For the normalized box, $\int_{\mathbb{R}} \phi_{i,n}(t)\,\mathrm{d}t = 2^{-i/2}$. A coefficient obtained by pairing with this box is therefore a scaled average, not the average itself. The reconstruction formula supplies the compensating factor.

6.4 Coefficients and local averages

At resolution i, acquire the normalized coefficients

$$a_i[n] := \langle f, \phi_{i,n} \rangle_{L^2} = 2^{i/2} \int_{I_{i,n}} f(u)\,\mathrm{d}u. \tag{6.5}$$

Reconstruction uses the same normalized functions:

$$f_i(t) := \sum_{n \in \mathbb{Z}} a_i[n]\phi_{i,n}(t).$$

If $t \in I_{i,n}$, the only surviving factor is $\phi_{i,n}(t) = 2^{i/2}$. Therefore

$$\boxed{f_i(t) = 2^{i/2} a_i[n] = 2^i \int_{I_{i,n}} f(u)\,\mathrm{d}u, \qquad t \in I_{i,n}.} \tag{6.6}$$

Since $|I_{i,n}| = 2^{-i}$, the last expression is precisely the cell average.

It is useful to distinguish the integral, the normalized coefficient, and the average. If $m_i[n]$ denotes the average value on a cell, then

$$\underbrace{\int_{I_{i,n}} f(u)\,\mathrm{d}u}_{\text{raw integral}} = 2^{-i} m_i[n], \qquad \underbrace{a_i[n]}_{\text{unit-energy coefficient}} = 2^{-i/2} m_i[n].$$

Multiplication by the reconstruction height $2^{i/2}$ returns $m_i[n]$. In particular, any cell on which f is constant with value b is reconstructed with that same value. Cells crossing a change in f may have a different average.

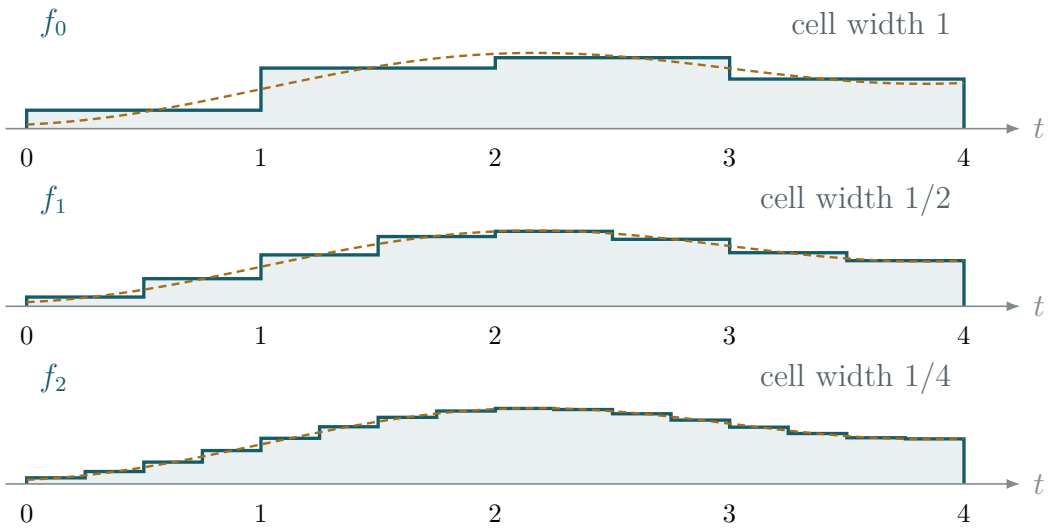


Figure 6.5: The same signal and averaging rule at successively finer resolutions. The plotted step heights are averages; their normalized expansion coefficients are smaller by the factor $2^{-i/2}$.

At a finer resolution, each unit interval contains more cells, so the approximation can retain more of the variation of the signal. The acquisition and reconstruction rules have not changed. We have only applied the same local averaging procedure at a smaller scale.

6.5 The detail between two resolutions

Increasing the resolution retains variation within each coarse cell. To identify this additional information, compare a fine approximation with its local average.

Consider one cell $I_{i,n}$, split into two children. Suppose the finer approximation has value α on the left child and β on the right. Its coarse average is

$$m = \frac{\alpha + \beta}{2}.$$

Subtracting the average gives two opposite residuals:

$$\alpha - m = \frac{\alpha - \beta}{2}, \qquad \beta - m = -\frac{\alpha - \beta}{2}.$$

The two fine values have separated into two kinds of information: their common average and their difference. The average remains when the resolution is reduced, while the opposite residuals record exactly what is lost. Adding these residuals back recovers the fine values.

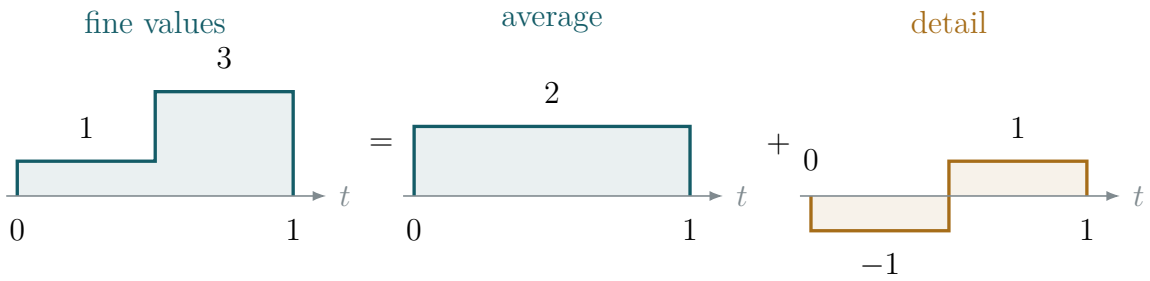


Figure 6.6: Two fine values, 1 and 3, become the average 2 and opposite details $-1, +1$. Adding the two right-hand plots returns the left-hand plot exactly.

The Haar wavelet. The pattern that describes this difference is

$$\psi(t) := \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t), \qquad \psi_{i,n}(t) := 2^{i/2}\psi(2^i t - n). \tag{6.7}$$

The function ψ is the Haar *mother wavelet*; ϕ is the Haar *scaling function*, also called the father wavelet. Direct calculation gives

$$\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = \frac{1}{2} - \frac{1}{2} = 0, \qquad \|\psi\|_{L^2}^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

The substitution from equation (6.4) also gives $\|\psi_{i,n}\|_{L^2} = 1$ and $\int_{\mathbb{R}} \psi_{i,n}(t)\,\mathrm{d}t = 0$.

Define the detail function $q_i := f_{i+1} - f_i$. On the chosen parent cell,

$$q_i(t) = \frac{\alpha - \beta}{2}\,\psi(2^i t - n) = 2^{-i/2}\frac{\alpha - \beta}{2}\,\psi_{i,n}(t). \tag{6.8}$$

Equal child averages give zero detail. These are changes relative to a resolution, not isolated Fourier frequencies.

6.6 A complete multiresolution decomposition

Take a signal supported on $[0, 1)$ with successive eighth-cell values

$$(1, 3, 5, 7, 7, 7, 5, 5).$$

It is already constant on each cell at resolution 3, so $f = f_3$. Repeated averaging gives

resolution	cell width	approximation values
3	1/8	$(1, 3, 5, 7, 7, 7, 5, 5)$
2	1/4	$(2, 6, 7, 5)$
1	1/2	$(4, 6)$
0	1	(5)

For example, $(1+3)/2 = 2$, $(5+7)/2 = 6$, and $(2+6)/2 = 4$. The difference at each stage records exactly what the averaging removes.

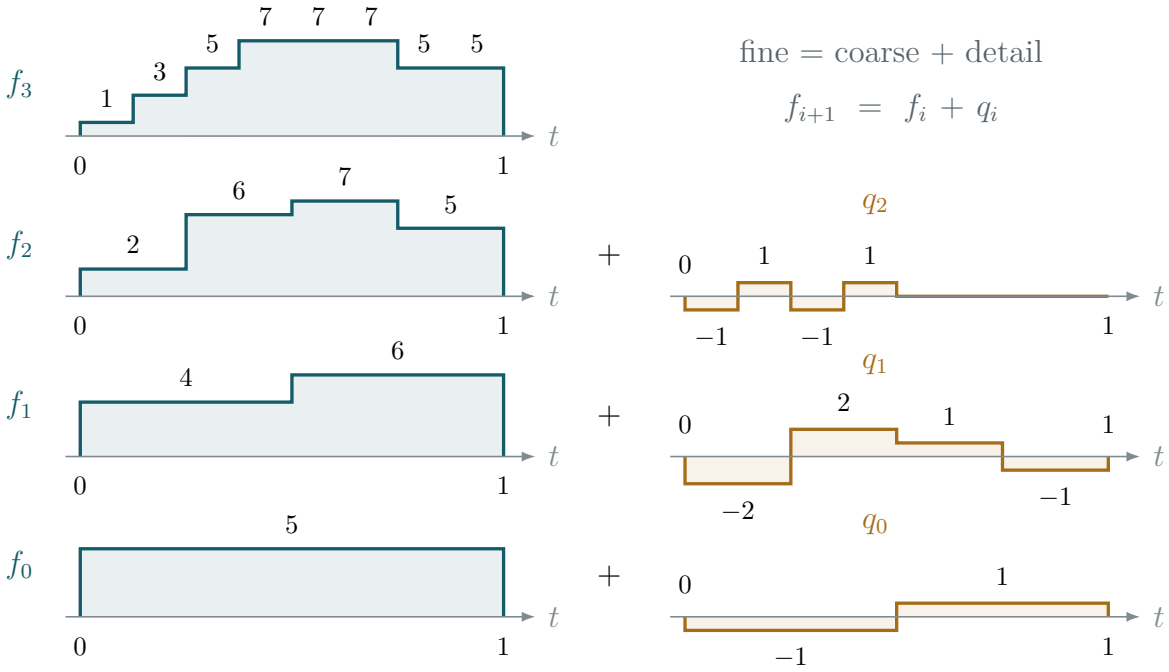


Figure 6.7: A three-level decomposition. All plots use the same vertical scale and are zero outside $[0, 1)$. Each detail has zero average on every parent cell.

Read each row as $f_{i+1} = f_i + q_i$. Repeated substitution gives

$$\boxed{f_3 = f_0 + q_0 + q_1 + q_2.} \tag{6.9}$$

The numbers in the plots are *function values*, not normalized basis coefficients. We calculate those coefficients after deriving the two-scale equations.

6.7 Approximation spaces and detail spaces

The example separates an approximation into averages and differences. We now describe all functions that can occur as either component, so that the same construction applies to any finite-energy signal.

At a fixed resolution i, define

$$V_i \coloneqq \overline{\operatorname{span}\{\phi_{i,n} : n \in \mathbb{Z}\}}^{L^2}, \tag{6.10}$$

$$W_i \coloneqq \overline{\operatorname{span}\{\psi_{i,n} : n \in \mathbb{Z}\}}^{L^2}. \tag{6.11}$$

The overline means *closure in the L^2 norm*: include not only finite linear combinations, but also signals that these combinations approximate arbitrarily well in energy. This allows finite-energy signals extending across infinitely many cells.

The space V_i consists of finite-energy functions constant on each $I_{i,n}$. The space W_i consists of finite-energy functions constant on each half-cell, with opposite values on the two halves of every parent cell.

Orthonormal functions at each resolution. Distinct cells have disjoint interiors and all the generators have unit energy. Hence

$$\langle \phi_{i,n}, \phi_{i,m} \rangle_{L^2} = \langle \psi_{i,n}, \psi_{i,m} \rangle_{L^2} = \delta[n-m].$$

Also $V_i \perp W_i$. For different cells the inner product vanishes by disjointness; on the same cell,

$$\langle \phi_{i,n}, \psi_{i,n} \rangle_{L^2} = 2^i \left(2^{-i-1} - 2^{-i-1}\right) = 0.$$

The notation $V_i \perp W_i$ means that every element of one space is orthogonal to every element of the other.

The cell-average approximation is a projection. Write $P_i f = f_i$. On each cell, equation (6.6) gives

$$\int_{I_{i,n}} (f(t) - f_i(t))\, \mathrm{d}t = 0.$$

Completing the square on a cell gives $\int_I |f-m|^2 = \int_I |f|^2 - |I||m|^2 \geq 0$, where m is its mean. Summing over cells shows that f_i has finite energy. The zero residual integrals make $f - f_i$ orthogonal to V_i. Thus, for $v \in V_i$,

$$\|f - v\|_{L^2}^2 = \|f - f_i\|_{L^2}^2 + \|f_i - v\|_{L^2}^2. \tag{6.12}$$

Therefore $P_i f$ is the unique best approximation in V_i. In particular,

$$\|f\|_{L^2}^2 = \|f - P_i f\|_{L^2}^2 + \|P_i f\|_{L^2}^2, \qquad \|P_i f\|_{L^2} \leq \|f\|_{L^2}.$$

For an arbitrary acquisition function, reconstruction with the same translates need not be a projection. Here it is a projection because the Haar boxes are orthonormal.

6.8 Nested spaces and the two-scale equations

A function constant on a cell is also constant on each of its two children. A Haar detail also has constant values on those children. Consequently

$$V_i \subset V_{i+1}, \qquad W_i \subset V_{i+1}.$$

In particular, both $\phi \in V_0$ and $\psi \in W_0$ can be built from the finer boxes in V_1.

Split the box. The two half-intervals partition $[0, 1)$, so

$$\phi(t) = \phi(2t) + \phi(2t - 1).$$

Since $\phi_{1,0}(t) = \sqrt{2}\phi(2t)$ and $\phi_{1,1}(t) = \sqrt{2}\phi(2t - 1)$,

$$\boxed{\phi(t) = \frac{1}{\sqrt{2}}\phi_{1,0}(t) + \frac{1}{\sqrt{2}}\phi_{1,1}(t).} \tag{6.13}$$

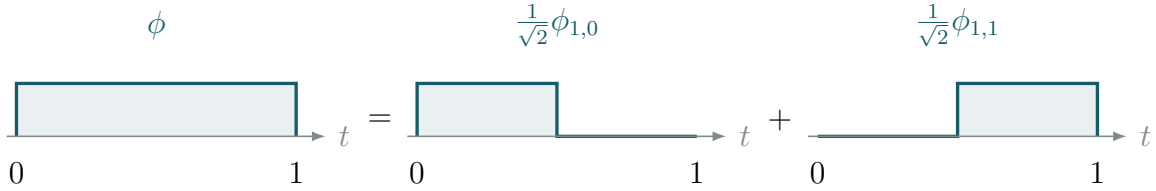


Figure 6.8: The scaling function is the sum of two weighted finer boxes. Each weighted term has height one on its own half-interval.

Split the wavelet. The right half has the opposite sign:

$$\psi(t) = \phi(2t) - \phi(2t - 1).$$

Therefore

$$\boxed{\psi(t) = \frac{1}{\sqrt{2}}\phi_{1,0}(t) - \frac{1}{\sqrt{2}}\phi_{1,1}(t).} \tag{6.14}$$

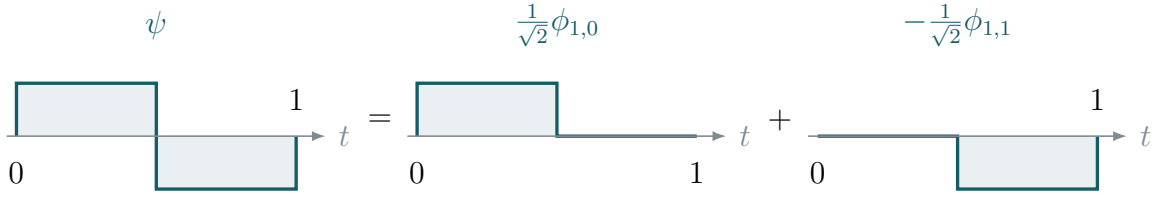


Figure 6.9: The wavelet uses the same two finer boxes with opposite coefficients. The negative term supplies the right-hand half of the Haar pattern.

These are the *two-scale equations*. Each expresses a function at one scale in terms of scaling functions at the next finer scale. The factors $1/\sqrt{2}$ are basis coefficients: they turn height-$\sqrt{2}$ boxes into height-one pieces.

6.9 The two-scale coefficients are filters

The coefficients in the two-scale equations form short discrete sequences. Their frequency responses explain why one sequence computes averages and the other computes differences. Write the equations in their filter form:

$$\phi(t) = \sum_{n\in\mathbb{Z}} g_0[n]\sqrt{2}\phi(2t-n), \tag{6.15}$$

$$\psi(t) = \sum_{n\in\mathbb{Z}} g_1[n]\sqrt{2}\phi(2t-n), \tag{6.16}$$

where

$$g_0[n] = \frac{\delta[n]+\delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n]-\delta[n-1]}{\sqrt{2}}.$$

Only the taps at $n = 0, 1$ are nonzero.

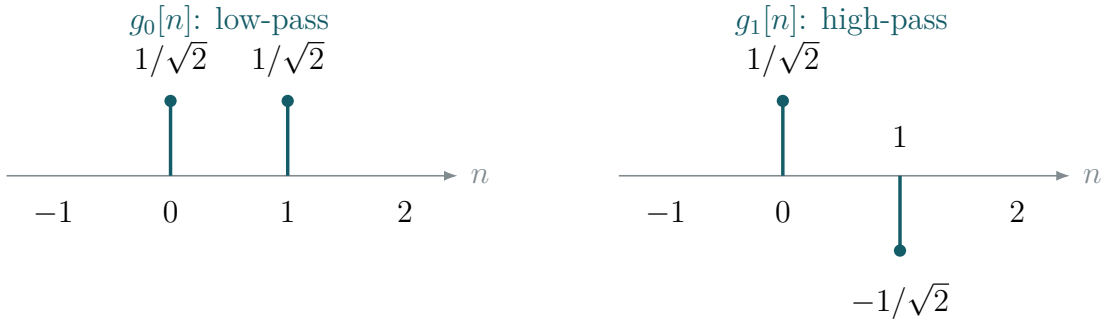


Figure 6.10: Both masks have energy one; their inner product is $1/2 - 1/2 = 0$.

Frequency responses (ω in radians per sample):

$$G_0(\mathrm{e}^{\mathrm{j}\omega}) = \frac{1+\mathrm{e}^{-\mathrm{j}\omega}}{\sqrt{2}} = \sqrt{2}\mathrm{e}^{-\mathrm{j}\omega/2}\cos(\omega/2),$$

$$G_1(\mathrm{e}^{\mathrm{j}\omega}) = \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\sqrt{2}} = \sqrt{2}\mathrm{j}\mathrm{e}^{-\mathrm{j}\omega/2}\sin(\omega/2).$$

The factorization is $1 \pm \mathrm{e}^{-\mathrm{j}\omega} = \mathrm{e}^{-\mathrm{j}\omega/2}(\mathrm{e}^{\mathrm{j}\omega/2} \pm \mathrm{e}^{-\mathrm{j}\omega/2})$. The sum in parentheses is $2\cos(\omega/2)$; the difference is $2\mathrm{j}\sin(\omega/2)$.

At $\omega = 0, \pi$, respectively, the responses are $(\sqrt{2}, 0)$ for g_0 and $(0, \sqrt{2})$ for g_1: an average rejects alternating signs, while a difference rejects constants.

6.10 An orthogonal direct sum

Proposition 6.1. At every integer resolution,

$$\boxed{V_{i+1} = V_i \oplus W_i.}$$

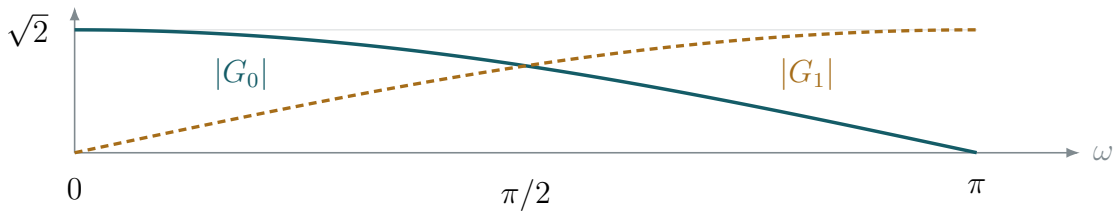


Figure 6.11: Magnitude responses on $[0, \pi]$. Their squared magnitudes add to two; neither filter has an ideal rectangular passband.

Every $v \in V_{i+1}$ has a unique decomposition $v = p + q$, with $p \in V_i$ and $q \in W_i$, and the two components are orthogonal.

Proof. On each parent cell, let α, β be the two values of v. Set

$$m = \frac{\alpha + \beta}{2}, \qquad r = \frac{\alpha - \beta}{2}.$$

Let $p = m$ on the whole parent cell, and let q take the values $r, -r$ on its two halves. Then $p \in V_i$, $q \in W_i$, and

$$m + r = \alpha, \qquad m - r = \beta.$$

These functions have finite energy: the identity

$$|m|^2 + |r|^2 = \frac{|\alpha|^2 + |\beta|^2}{2}$$

shows that their energies add to the energy of v after multiplication by the parent-cell width and summation over cells.

We already proved $V_i \perp W_i$. If a function u belongs to both spaces, it is orthogonal to itself, so $\|u\|_{L^2}^2 = 0$. Thus $V_i \cap W_i = \{0\}$. Subtracting two decompositions would put their difference in this intersection, proving uniqueness. □

The symbol $\oplus$ records that the decomposition is unique; here the two summands are also orthogonal. Thus the extra freedom in V_{i+1} consists precisely of the detail space W_i. Applying this decomposition to the approximations of a fixed signal gives

$$P_{i+1} f = P_i f + q_i, \qquad q_i = P_{i+1} f - P_i f \in W_i. \tag{6.17}$$

Moreover, q_i is the orthogonal projection of f onto W_i. Indeed, $f - q_i = (f - P_{i+1} f) + P_i f$: the first term is orthogonal to V_{i+1}, and the second to W_i.

Compute the coefficients without resampling the function. Define

$$a_i[n] = \langle f, \phi_{i,n} \rangle_{L^2}, \qquad d_i[n] \coloneqq \langle f, \psi_{i,n} \rangle_{L^2}.$$

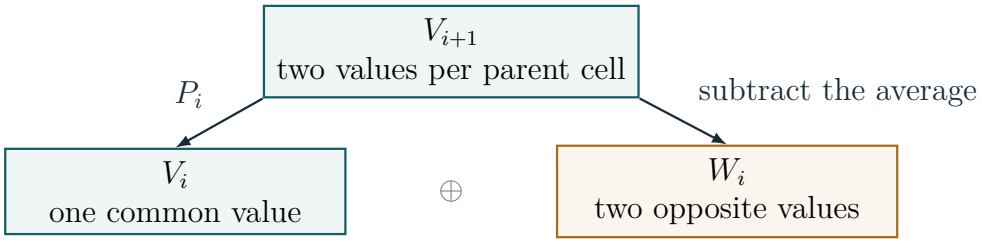


Figure 6.12: Averages and details occupy complementary, orthogonal subspaces of the next finer approximation space.

Dilating and translating equations (6.13) and (6.14) gives

$$\phi_{i,n} = \frac{\phi_{i+1,2n} + \phi_{i+1,2n+1}}{\sqrt{2}},$$
$$\psi_{i,n} = \frac{\phi_{i+1,2n} - \phi_{i+1,2n+1}}{\sqrt{2}}.$$

For example, substitute $2^i t - n$ for t in the box equation and multiply by $2^{i/2}$; each child has normalization $2^{(i+1)/2}$. Taking inner products, the coefficients $1/\sqrt{2}$ are real, so conjugation does not change them:

$$\boxed{\begin{aligned} a_i[n] &= \frac{a_{i+1}[2n] + a_{i+1}[2n+1]}{\sqrt{2}}, \\ d_i[n] &= \frac{a_{i+1}[2n] - a_{i+1}[2n+1]}{\sqrt{2}}. \end{aligned}} \tag{6.18}$$

Adding and subtracting these equations recovers the fine coefficients:

$$a_{i+1}[2n] = \frac{a_i[n] + d_i[n]}{\sqrt{2}}, \qquad a_{i+1}[2n+1] = \frac{a_i[n] - d_i[n]}{\sqrt{2}}. \tag{6.19}$$

The detail reconstructs as $q_i = \sum_n d_i[n]\psi_{i,n}$. Keeping both coefficient outputs gives perfect reconstruction.

Return to the eight-cell example. A cell value at resolution i must be multiplied by $2^{-i/2}$ to obtain its coefficient. Applying equation (6.18),

i	a_i	d_i
3	$(1,3,5,7,7,7,5,5)/\sqrt{8}$	—
2	$(1,3,7/2,5/2)$	$(-1/2,-1/2,0,0)$
1	$(2\sqrt{2},3\sqrt{2})$	$(-\sqrt{2},1/\sqrt{2})$
0	(5)	(-1)

For the first pair, $\sqrt{8}\sqrt{2} = 4$, so $a_2[0] = (1+3)/4 = 1$ and $d_2[0] = (1-3)/4 = -1/2$. The complete Haar expansion is

$$f = 5\phi - \psi - \sqrt{2}\psi_{1,0} + \frac{1}{\sqrt{2}}\psi_{1,1} - \frac{1}{2}\psi_{2,0} - \frac{1}{2}\psi_{2,1}. \tag{6.20}$$

All other coefficients at these resolutions are zero. The filter operations rearrange the same information; discarding details is the information-losing step. These analysis and synthesis operations form a perfect-reconstruction filter bank and lead to the discrete wavelet transform (DWT). Just as the discrete Fourier transform provides a discrete-domain form of Fourier analysis, the DWT gives a discrete-domain procedure for working with the multiresolution representation.

6.11 The Haar multiresolution analysis

The approximation spaces form a nested family:

$$\cdots \subset V_{-2} \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots \subset L^2(\mathbb{R}).$$

Each is a dilated version of V_0: $v \in V_i$ exactly when $v(2^{-i}\,\cdot) \in V_0$. All grids use the same origin, with finer cells as i increases.

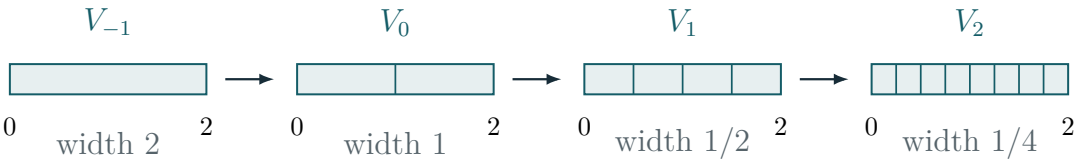


Figure 6.13: The same displayed interval, $[0, 2)$, at four resolutions. The spaces themselves are defined on the entire real line; only the cell width changes.

The two limiting statements in the multiresolution picture have precise meanings:

$$\boxed{\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{\,L^2} = L^2(\mathbb{R}), \qquad \bigcap_{i\in\mathbb{Z}} V_i = \{0\}.} \tag{6.21}$$

The first says that sufficiently fine piecewise-constant approximations can achieve arbitrarily small energy error. The second says that no nonzero finite-energy signal can be constant on the cells at *every* scale.

Finer resolutions recover the signal. For a signal p continuous on a closed, bounded interval, averaging over a cell I cannot change it by more than its variation inside that cell:

$$\left| p(t) - \frac{1}{|I|}\int_I p(u)\,\mathrm{d}u \right| \le \sup_{u\in I} |p(t) - p(u)|, \qquad t \in I.$$

As the cell width tends to zero, this bound tends to zero uniformly on that interval. Hence the energy error tends to zero as well.

To pass to a general $f \in L^2(\mathbb{R})$, first discard tails of arbitrarily small energy. On each of the remaining finitely many unit intervals, approximate f by a finite Fourier sum, using the Fourier-series theorem from Lecture 5. Call the resulting piecewise-smooth signal p; its support is bounded. Fine dyadic cells do not cross the integer boundaries, so the preceding argument applies on each interval. Since P_i cannot increase the L^2 norm,

$$\|f - P_i f\|_{L^2} \le 2\|f - p\|_{L^2} + \|p - P_i p\|_{L^2}.$$

First make $\|f - p\|_{L^2}$ arbitrarily small, then take i large enough to make the last term small. Thus $\|f - P_i f\|_{L^2} \to 0$, proving the first part of equation (6.21).

Coarser resolutions leave no finite-energy background. Suppose v belongs to every V_i. At resolution $i = -m$, it is constant on $[0, 2^m)$ and on $[-2^m, 0)$. As m grows, these intervals exhaust the two half-lines. Their overlaps force the constants to agree on each half-line. A nonzero constant on a half-line has infinite energy. Hence both constants must be zero, proving the second part of equation (6.21).

The coarse projections themselves also tend to zero. For a finite-energy signal p with bounded support, write $L = 2^{-i}$. Its cell averages satisfy

$$\begin{aligned} \|P_i p\|_{L^2}^2 &= \frac{1}{L} \sum_n \left| \int_{I_{i,n}} p(t)\, \mathrm{d}t \right|^2 \\ &\le \frac{1}{L} \left(\int_{\mathbb{R}} |p(t)|\, \mathrm{d}t \right)^2 \longrightarrow 0 \qquad (i \to -\infty). \end{aligned}$$

Here the inequality follows from $|\int_I p| \le \int_I |p|$ and $\sum_n u_n^2 \le (\sum_n u_n)^2$ for $u_n \ge 0$. For general f, approximate by such a p and use

$$\|P_i f\|_{L^2} \le \|P_i(f - p)\|_{L^2} + \|P_i p\|_{L^2} \le \|f - p\|_{L^2} + \|P_i p\|_{L^2}.$$

The whole-line setting matters: unlike on $[0, 1]$, the constant function is not a nonzero finite-energy background on $\mathbb{R}$.

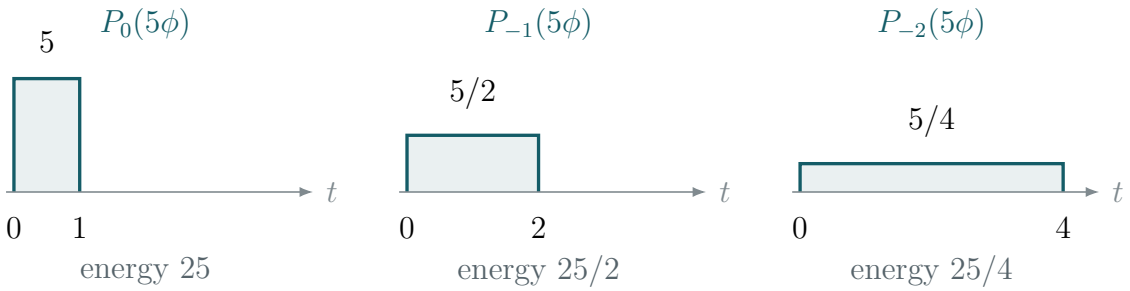


Figure 6.14: Coarsening the example's background 5ϕ. Its integral stays equal to five, but its amplitude and energy decrease as that integral is spread over longer intervals.

Together, the nested scales, the two limits, and the orthonormal translates of ϕ give the *Haar multiresolution analysis*, abbreviated MRA. It is generated by one scaling function and one rule for changing scale. The detail spaces are determined by the approximation spaces: W_i consists exactly of the functions in V_{i+1} that are orthogonal to V_i. In the next lecture, we retain this structure while allowing scaling functions other than the Haar box.